Engineering Mechanics 1 · Unit_01 · UNIT 1

DRAFT — UE26ME142A · Unit_01 · UNIT 1 · UE26ME142A Unit 1 Class 4 AVSummary

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · UE26ME142A Unit 1 Class 4 AVSummary
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

UE26ME142A Unit 1 Class 4 AVSummary

1 UE26ME142A Unit 1 Class 4 AVSummary

Study focus: UE26ME142A_Unit 1_Class 4_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000106 · EV0000107

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I am Dr.

Key details

- SV Venkitesh, Civil Engineering, Department of Science and Humanities, PS University.
- In the last class, that is class number six, we had discussed several problems and how do we solve them by both scalar approach as well as vector approach and they were solved problems.
- The statement of the problem is like this.
- Two forces are applied to the construction bracket as shown.
- Determine the angle cheetah, which makes the resultant of the two forces vertical.
- Determine the magnitude R of the resultant.
- And they have also asked you to determine the magnitude of the resultant R.
- No down, this is the angle made by the force F1.
- With respect to X, okay.
- The force F1 is in the plane of X and Y only.
- In the XY plane itself.
- Now to get the result of these two forces, first thing that I'm going to do is I'll resolve both the forces F1 as well as F2 along its components, that is along x-axis and y axis.
- F1x and F1Y.
- I get F1x as F1.
- cos 70 degrees f1 y as f1 sinilarly for this f2 x will be equal to f2 cos theta and f2 y will be f2 sine theta no adding up of.
- The components along x axis, okay, that is the component of f1 along x, that is f1x, and the component of f2 along x, which is f2x, if I add these two, that should be.

- equal to zero.
- Why should that be equal to zero?
- Because what are we supposed to find out is the angle theta which makes the resultant to be vertical.
- It should be 0.
- Because F1 is acting along X, it's positive.

Example or application from the lecture

- From the diagram, what is that we need to look at is they have given you the magnitude of the force F1, which is 800 newtons, and they have also given you what is the direction of this force F1 with respect to the negative x axis, which makes an angle of 70 degrees.
- Next, they have also given you the magnitude of the force F2, which is equal to 425 newtons, and the angle theta which we have to determine is the angle made by the force F2 with respect to x axis again.
- Even though you see that the diagram that is given, the bracket that is given is three-dimensional, but the forces acting is in the same plane that is XY plane.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

Core explanation · 05:00–09:59

The component of F1 acting along X is positive.

Key details

- The component of the force.
- Along X of F2 is negative because the sense of it is opposite to X.
- By solving it, we are going to get the value of theta which is 49.92.
- Then, so that is what we have to find out in the first case, theta.
- Theta is what we have got.
- The second case, they have also asked you to determine the magnitude of R.
- Find out the summation of that.
- That will be equal to the resultant force.
- And the force, the resultant force will be acting along y only, so you get that value.
- Both F1 Y and F2 Y both of them are acting downwards, so it will be negative.
- The component of F2 along y will be minus F2 into sin theta, which is equal to minus r y, which is equal to minus r.
- That is the resultant.
- I hope all of you have understood this problem.
- Now we'll move on to the next problem.
- The next problem that I've chosen is 2 bar 17.
- The statement of the problem goes like this.
- On the TB.
- Read out from the beginning.
- The two forces shown at in the XY plane of the T beam cross section.

- A line of action that lies 15 degrees above the negative x axis.
- What is this?
- The resultant will be acting somewhere here.
- Resultant, of course, the point of application will be the same as the point of application of the two forces.
- Or F1, which is which we have to find out, or rather, what is the direction of the force F1, which makes an angle of 30 degrees with the y-axis, then what is the magnitude of F2, which is 2.8 kilonewtons.
- And the angle made by F2 with respect to x axis is theta.
- We have to determine the magnitude of F1 and we have to determine the direction of F2.
- That is with respect to x axis.
- F2 is acting towards the origin, the magnitude of that is 2.8, and it makes an angle of theta with respect to the positive x-axis.
- Excuse me, F1 is also acting towards the origin.
- It makes an angle of 30 degrees with the negative y axis.

Example or application from the lecture

- Obviously, this is acting downwards, only the magnitude is given here, and of course it is acting downwards.
- The two forces shown act in the XY plane, they have already given that it is acting in the XY plane.
- Even the diagram that is given is three-dimensional, but since they are told that the forces acting is in the XY plane, we can treat this as a two-dimensional force system.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000110 · EV0000111 · EV0000112 · +1 source units in pack.json

Core explanation · 10:00–14:59

Magnitude is 3.5 kilonton.

Key details

- Alpha is the angle made by the resultant with respect to the negative x-axis of the state is above, that is 15 degrees.
- What are we supposed to find?
- Is the magnitude of F1 and the angle theta of the force F2 with respect to x-axis.
- F2 also moving away and the result end also.
- How is it 45?
- 90 minus 30 minus 15.
- will be forty five degrees.
- Using the law of triangle of forces, so F1, F2 taken in order.
- The magnitude of the force F2 as 2.8, and we are supposed to find out the magnitude of F1.
- We know the angle made by F1 with the resultant R that is 45 degrees.
- We can also determine.

- What is the angle made by the force F2 and the resultant which will be equal to theta plus 15?
- How is it?
- It's theta plus 15 here.
- When I put it here the this would be the alternate angle for this which will also be equal to theta plus fifteen.
- It states that 3.5 divided by sine of the sine of this angle, that is sine of 120 minus theta, should be equal to.
- Magnitude of this force which is two point eight, opposite angle as forty five, so sine of forty five should also be equal to this force F one.
- Angle opposite to that is this which is.
- Equal to theta plus 45 sine of theta plus 45.
- And 3.5 sine 120 minus theta.
- If I take these two equations, I can rewrite this equation as sin of 120 minus theta equals 3.5 sine 45 divided by 2.8, from which I can find out the value of theta.
- By simplifying it.
- Once I get the value of theta, I can take these two equations, okay, substitute the value of theta here, and then find out the value of f1.
- I have not gone for resolution of forces F1 and F2, because when you go for resolution of forces, the solution will be a lengthier one.
- The law of triangle of forces or using the sine rule.
- Law of triangle of forces is nothing but which we have got it from sine rule.
- I've taken one more problem, 2 bar 22.
- The statement of the problem goes like this.
- A sheet of an experimental composite is subjected to a simple tension test.
- To determine its strength along a particular direction, the composite is reinforced by the Keplar fibers shore.
- And a close up showing the direction of the applied tension force F in relation to the fiber directions at point A is shown.
- If the magnitude of F is 2.5 kilonewt, determine the components FA.

Example or application from the lecture

- Let's look at the solution.
- Okay, let's go on to the next problem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000114 · EV0000115 · EV0000116 · +1 source units in pack.json

Core explanation · 15:00–19:59

And F B of the force F along the oblique axis, A and B also determine the projection P A and P B of the force F onto the A and B axis.

Key details

- If you take a small enlarged view of the intersection of the fibers, it is got to look like this.

- What are we supposed to find?
- The component of the force F along axis A and B, as well as the projection of the force along A and B.
- That is, this is the axis B and the force F is acting along X.
- I hope it is clear.
- we have got the angles, so now using the sign rule to get the components.
- Components I have told you will follow law of triangle of forces or law of parallelogram of forces.
- The law of parallel of forces or the law of triangle of forces is derived from sine rule, so we can use sine rule to get it.
- I can write it as FA, which is the force along A divided by the sine of the angle between the other two.
- Divided by the sine of the angle between the other two, which is x-axis and a that is sine of 40, which should be equal to f, which is the fourth sine of the angle between the other two, which is the total angle between A and B, which is sine 50.
- I can get the value of FA and FP.
- The units of the forces are important.
- Kilonewtons, so you will have the components of the forces also in kilonewtons.
- Next coming to the projection.
- Projection is the cause component of the force.
- Here you can see the projection, projection of PA, that is projection of the force F.
- On A, how do I get?
- If this is A, I should draw a line from the edge of the force to the axis A, such that this line would make an angle of 90 degrees.
- F cos 40 would give me the projection of the force on A.

Example or application from the lecture

- What is given is you have a diagram indicating the composite sheet to which a force F is applied, a tensile force F is applied.
- They have also given you what is the angles made by these fibers with respect to the axis.
- The tensile force is given as 2.5.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000118 · EV0000119 · EV0000120 · +1 source units in pack.json

Core explanation · 20:00–24:59

That is what is written here.

Key details

- F cost 40.
- This angle is 90 degrees, so this forms a.
- Okay, so we will stop here because we are running short of time.
- We'll continue in the next class.
- My name is Dr.

- SV Venpej from Civil Engineering Department of Science and Humanities PS University.
- We'll meet in the next class, that is class number eight.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000122 · EV0000123 · EV0000124 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE26ME142A_Unit 1_Class 4_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/004_UE26ME142A_Unit_1_Class_4_AVSummary__93feea637b67e79c.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 4_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/004_UE26ME142A_Unit_1_Class_4_AVSummary__93feea637b67e79c.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 4_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/004_UE26ME142A_Unit_1_Class_4_AVSummary__93feea637b67e79c.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 4_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/004_UE26ME142A_Unit_1_Class_4_AVSummary__93feea637b67e79c.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.slides.md have no readable `.slides.md` text; their original page images are included.
4. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md have no readable `.slides.md` text; their original page images are included.
5. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md have no readable `.slides.md` text; their original page images are included.
7. 3 rendered pages in 004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.slides.md have no readable `.slides.md` text; their original page images are included.
8. 3 rendered pages in 005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.md have no readable `.slides.md` text; their original page images are included.
9. 2 rendered pages in 006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md have no readable `.slides.md` text; their original page images are included.
10. 3 rendered pages in 008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.md have no readable `.slides.md` text; their original page images are included.
12. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.slides.md have no readable `.slides.md` text; their original page images are included.

13. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.slides.md have no readable `.slides.md` text; their original page images are included.
14. 2 rendered pages in 011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.md have no readable `.slides.md` text; their original page images are included.
16. 4 rendered pages in 012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.slides.md have no readable `.slides.md` text; their original page images are included.
17. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
18. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
19. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
20. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
21. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00026	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00027	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00028	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00029	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

C Revision and retrieval practice